

Chapter #4

Uses-Requirement Analysis

The process of gathering information about the needs and requirements of the users of the data warehouse

⇒ After investigating the available information resources in the data source analysis and reconciliation phase, approach end users again to get more accurate information.

→ The user requirement analysis phase attempts to collect end user needs for data mart applications and usage.

→ Importance: This phase holds a strategic significance for designing data marts, because it influences every decision made during the project.

→ it plays necessary role in:

- Defining Conceptual schemata of data
- Data staging design
- Requirements for data analysis applications
- Data warehouse system architecture
- Instructions for data warehouse system maintenance

• Upgrades

Data come from this Phase:

Challenges:

→ The user-requirement analysis phase delivers ambiguous, incomplete and short lived requirements in data warehousing projects for ^{long-term nature:}

- These are long-term projects and it is difficult to get and arrange every requirement from the beginning.
- ^{Complexity of decision making phase:} The information requirements of data warehouse applications are difficult to explain because decision-making process has very flexible structure.

^{Difficulty in explaining information:}
• Decision making process requirements frequently make reference to a set of information that is not available in suitable format and needs to be derived from data sources.

Approaches to user-requirement analysis phase:

⇒ There are two different approaches:
~~User-requirement analysis~~

1. Informal approach
2. Formal approach

1. Informal approach:

- Requires glossaries to support design in the conceptual design phase.
- Properly fits into a data-driven design framework that emphasizes deriving data model schemata from operational source schemata.

2. Formal approach:

- Based on Tropos formalism for user-requirement analysis.
- Tropos formalism helps in designing system based on the needs and goals of users.
- This approach emphasizes that system design should be directly derived from the user's requirements.
- It integrates user requirements with practical limitations, like existing rules or processes to create a balanced design.

Ways of Requirement elicitation

1: Interviews

1. Business users:

⇒ The primary sources of requirements for data warehouse projects are the end users, also known as business users.

2. Challenges in interviews:

⇒ Interviews can be difficult due to language differences between designers and end users, and end-users' limited understanding of data warehousing.

3. Accurate requirements:

⇒ Accurate requirements are essential for ensuring end user satisfaction & the successful implementation of the data warehouse.

4. Technical know-how:

⇒ Information system administrators and data processing center staffs can provide valuable insights into technical constraints.

5. Constraints:

⇒ Constraints on data warehouse design, such as performance, scalability

and security must be considered during the requirement analysis phase.

Interview methods:

⇒ Two basic methods for conducting user requirement analysis are:

- Interviews
- Requirement analysis

1: Interviews

⇒ Can be conducted with individual users or small groups and offers the advantage of active participation and detailed specifications.

2: Facilitated sessions:

⇒ It involves large, heterogeneous groups lead by facilitators, who are responsible for setting up common language for all interviewees.

⇒ Can be helpful for encouraging creative brainstorming and setting general priorities.

Preliminary activities:

⇒ The main preliminary activities involved in interviews are:

1. Pre-interview research:

- Conduct research on the business issues and the specific application domain to raise awareness among interviewees.

- Also analyze previous data-warehousing projects to understand why they were unsuccessful.

2: Interview selection:

- Select a representative sample of end-users to obtain a wide range of information on the analysis needs in the data-most specific-application field.

3: Interview question development:

- Develop guidelines for interviews, but they should vary by interview job functions and levels.

4: Interview scheduling:

- Schedule the first meeting with the project sponsor to get their support and involvement.

5: Interviewee Preparation:

- Clarify interview goals and plans with all participants.

Types of questions during interview:

- Combination of open-ended, closed-questions and evidential questions to gather information.

1. Open-ended questions:

- Allow interviewees to express their opinions and ideas freely.

2. closed-ended questions:

- Focus the interview on specific topics and obtain direct answers.

3. Evidential questions:

- Ask interviewees to provide examples or evidence to support their answers.

Advantages of open-ended questions:

- Provide rich and detail information
- Allow interviewees express ideas freely.

Disadvantages of OE questions:

- Time-consuming
- Lead to digressions

Advantages of closed questions:

- Shorten the interview time
- Focus interviewers on relevant topics
- Make answers comparable

Disadvantages of closed questions:

- Can sound boring
- Limit the interviewee's freedom
- Assume that the interviewee knows the relevant topics.

Advantages of evidential questions:

- Allow the interviewee to learn about the interviewee's experience and knowledge.

Disadvantages of evidential questions:

- Can be time-consuming
- May not be relevant to the interview if the interview lacks experience.
- Can make interviewees nervous because the questions are probing.

Types of interviews

→ There are two types of approaches for doing interviews:

- Pyramid-Shaped
- Funnel-Shaped

1. Pyramid-Shaped interviews:

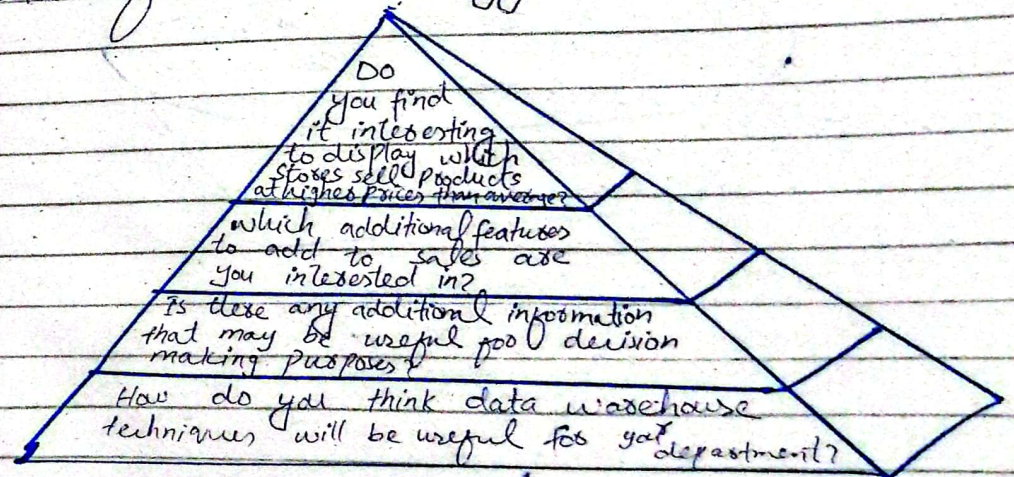
- it is an inductive approach: start with detailed questions and gradually become more general.
- Usually with closed questions.

Advantages:

overcomes interview hesitation and allows the interviewee to avoid getting deeply involved in the interview.

Example:

start with specific questions about data marts and then ask more general questions about decision-making processes and information technology.



2. Funnel-shaped interviews:

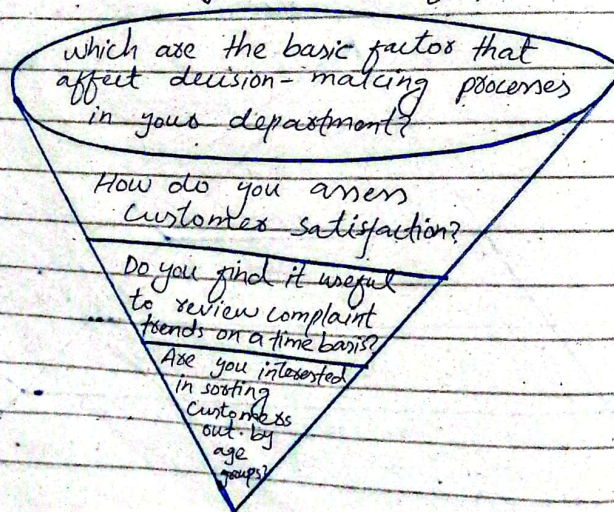
- Deductive approach.
- Start with general questions and gradually become more specific.

Advantages:

Useful if interviewees are excited or too differential, as general questions don't have wrong answers and can make interviewees feel more comfortable.

Example:

- Start with general questions about the company goals.
- Then ask more specific questions about data analysis and information needs.



2. Glossary-based Requirement Analysis:

Based on ↓
facts
→ preliminary workload.

→ A method of gathering information from users about what they need from the data warehouse. This method involves creating a glossary of terms that describe the data and how it should be used.

→ It is informal approach suitable when using data driven approach.

→ In 2001, Lechtenbörger proposed user-requirement analysis approach based on glossaries.

Glossaries:

- These are like dictionaries that define the specific terms and concepts related to the data warehouse.

Derivation table:

- This table shows how the data in the data warehouse will be derived from the original data sources.

Usage tables:

- This table explains how the data will be used for analysis and reporting.

→ it links textual description to attributes and specifies attribute roles such as analysis dimension and measure.

⇒ Both tables should list the attributes for data modeling.

Structure tables

- This table defines how the data will be organized and structured within the data warehouse.
- Specifies whether the attributes should be modeled as dimensions or attributes linked to dimensions or measure.

(i) Facts

→ Facts are fundamental units of data in a data warehouse.

→ They represent measurable events or occurrences that are the primary focus of analysis.

→ Facts are the basic units of information that users will analyze.

Dynamic feature: Facts should be able to change over time (e.g. sales figures etc.)

Relevance: A fact may be relevant for one data warehouse but not for another.

uses interest: The choice of facts depends on what users want to analyze.

Data sources: The fact should be based on the data available in the original data sources.

Example of fact in various application fields, along with various data masts:

1. In Healthcare application field, under the Division data mast, some typical facts include admissions, discharges, transfers, surgical operations, diagnosis and prescription. These facts can be used to analyze patient flow, treatment effectiveness and resource utilization within a healthcare facility.

2. In Business manufacturing, under Supplier data mast, some typical facts are purchases, stock inventory and distribution.

- Under demand management data mast some facts are sales, invoices, orders, shipments and complaints.

Fact granularity:

Refers to level of detail for data to be represented in a data mart.

For example: Patient data can be at a granular level of individual medical records or aggregated to a higher level, like monthly patient visits.

Trade-offs:

- Choosing a granularity level involves balancing flexibility and performance.
- Higher granularity allows for more detailed analysis but can be computationally expensive, while lower granularity might be faster to process but less flexible.

Important points:

- one of the most important features of data warehouse is representation of historical data. That help to track how data changes over time.
- historical interval: Every fact needs a historical interval to be defined. means each fact should have a specific time range associated with it.

→ The value of this interval typically range from three to five years.

Example of user - Requirement Glossary:

→ Requirement glossary table created by data mart designers focused on sales and inventory data.

Fact	Dimensions	measure	Historical interval
STOCK, INVENTORY	Product, date, warehouse	stocked quantity	1 Year
SALES	Product, date, store	Sold quantity, receipt, discount count	5 Years
ORDER LINES	Product, date, supplier	ordered quantity, receipt, discount	3 Years

(ii) Preliminary Workload

→ This involves identifying the facts, dimensions, and measures that will be included in the data warehouse. Basically, collect requirements on analysis measures at data mart level.

→ Fact identification:

- Primary measurements within data warehouse.

Dimension & measure identification:

- it involves identifying attributes that describe facts and measures that are quantifiable values associated with facts.

Workload

- Workload can be expressed in natural language that will help the designer to overcome the doubts about fact granularity and relevant measures.

Workload refinement

- Do after, aim of workload refinement is to validate conceptual schema created by designers.
- Detail information is collected after the conceptual design phase.

Example of Preliminary workload

Fact	Query
Stock INVENTORY	what is [Average quantity? availability of products] in data warehouse
SALES	what are { Total sales? Daily sales? Sales by product type or stores

Order Lines	what is { Order details? Total orders? Order amount?
-------------	--

⇒ The **STOCK INVENTORY** fact is concerned:

- The data should be organized by product type and time (day, week, month)
- The quantity of products in stock should be calculated by adding up the quantities in different warehouses.
- Average and maximum quantities are calculated along the time.

⇒ The **SALES** fact is concerned:

- The data should be organized by product type, product category, store and time (day, week, year).
- Also calculate average price per unit sold.

- The total quantity sold should be calculated by adding up the quantities sold in different stores and over time.

⇒ For **ORDER LINES** same as **STOCK INVENTORY**

Importance of:

(i) Hierarchies:

- way to organize data in levels like tree structure.
- Ensures attributes are consistent in different parts
- Hierarchies build on dimensions.

(ii) Aggregations:

- Calculations that summarize data.
- Choice of aggregation operators can significantly affect the logical and conceptual design of data warehouse.

⇒ In short, while designing a data warehouse, considering both hierarchies and aggregations to ensure that the data can be effectively analysed and used to answer business questions.

3: Goal oriented Requirement Analysis

→ This methodology focuses on understanding the goals of decision-makers in an organization rather than just collecting a list of requirements.

Abstraction sheets:

→ These sheets are used to analyze goals separately and consider how they relate to enterprise-specific activities.

→ Goal oriented approach based on:

Troops formalism:

→ Framework used to represent user requirements and includes both decision-making modeling and organizational modeling.

Organizational analysis:

→ Focuses on the stakeholders and actors involved in managing the organization.

Fact, dimension & measure analysis:

→ Involves identifying:

- facts dimensions measures that are relevant to the goals and organizational structure.

3-1: Introduction to TROOPS

- Developed by Boesicani at 2004
- Based on modeling framework Yu 1995
- Software designing methodology
- Focuses on understanding the goals of people and systems and how they interact.

Two main features:

(i) Agent-oriented methodology:

Troops uses agents, which are like actors or objects, to represent people and systems.

(ii) Goal-based approach:

Troops emphasizes understanding the goals of agents and how they work together to achieve those goals.

- Troops manages four software development stages:

1. Early requirement analysis:

- This phase focuses on understanding the goals of agents and how they work together to achieve those goals.

2. ^{Design} Early requirement analysis:

- This phase focuses on defining the system functions and features.

3. Architecture design:

- This phase involves designing the overall structure of the system.

4. Working plans:

- This phase involves creating a detailed plan for each software module.

Notations of Troops

Some notations of Troops methodology that applies to requirement analysis for data models:

1) Actors:

- Represent people, software agents involved in a system.
- Business stakeholders.
- Circles are used to graphically represent the actors.

2) Strategic dependencies:

- These are agreements between actors about their goals, tasks or resources.
- Represented by ovals.

3. Actor diagram:

- This diagram shows how actors interact with each other based on their strategic dependencies.
- Used to describe how actors depend on each other.

4. Rationale diagram:

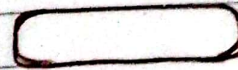
- This diagram shows the relationships between actors and their goals.
- Logical relationships shows.
- it looks like a bubble.
- Goals are the objectives that actors want to achieve.
- Goals broken down into smaller subgoals using AND Semantics (every subgoal must be reached) or the OR Semantics (any one of these subgoals must be reached).
- Positive or negative contributions blw goals are represented by arrows.
- That indicated one goal must be achieved before another.

- Contribution Show how one goal can help or hinder another goal.
- The +, ++, - and -- marks are used to rate the contributions.

Graphical Notations



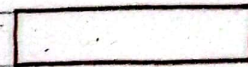
Actor



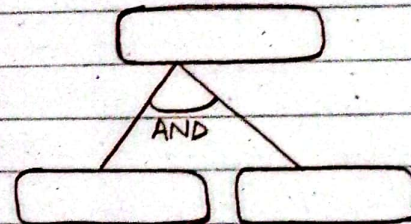
Goal



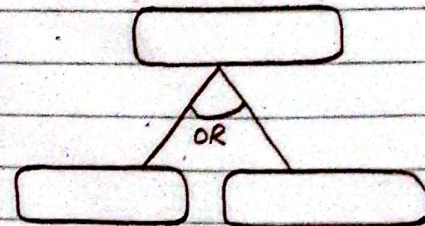
Dependency



Fact



AND-decomposition



OR-decomposition

⇒ To apply troops approach to data warehouse following concepts should be considered:

Facts:

→ From stand point of organizational modeling:

- a fact models a group of events taking place when an actor reaches a goal.

→ From standpoint of decision-making modeling:

- Facts are potential analysis items.

→ Rectangles connected to goals graphically mark facts in diagrams.

Attributes:

→ Field that is a given value to record a fact after reaching a goal.

→ Small diamonds connected to goals graphically mark facts in diagrams.

Dimensions:

→ Property of fact.

→ Help decision makers gain different perspectives on facts to reach analysis of goal.

→ Small circles connected to goals graphically mark dimensions in diagrams.

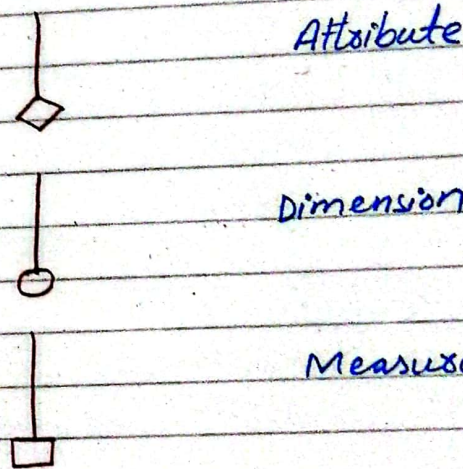
Measures:

→ Numerical property of fact.

→ Specifies quantitative feature of a fact.

→ Small squares connected to goal graphically mark measures in diagrams.

Graphical Notations



3-2: Organizational Modeling

→ Process used to understand and analyze an organization's structure, goals and relationships between different stakeholders.

→ To model organizations, a three phase process is follow.

→ Each phase consists of an iterative process whose inputs are the diagrams resulting from the previous step.

→ Three phases are as follows:

1. Goal analysis:

- Actor and rationale diagrams are created. These diagrams help understand the goals and relationships of different people or groups involved in the organization.

2. Fault Analysis:

- In second phase, more information is added to the rationale diagrams.
- This information is about faults that are happening in the organization.

3. Attribute Analysis:

- In third phase, rationale diagrams are extended further by adding more information.
- In this phase, specific characteristics or qualities of faults are specify.

3.2.1: Goal Analysis:

→ Once goals are analyzed, then actor diagram is created and rationale diagram.

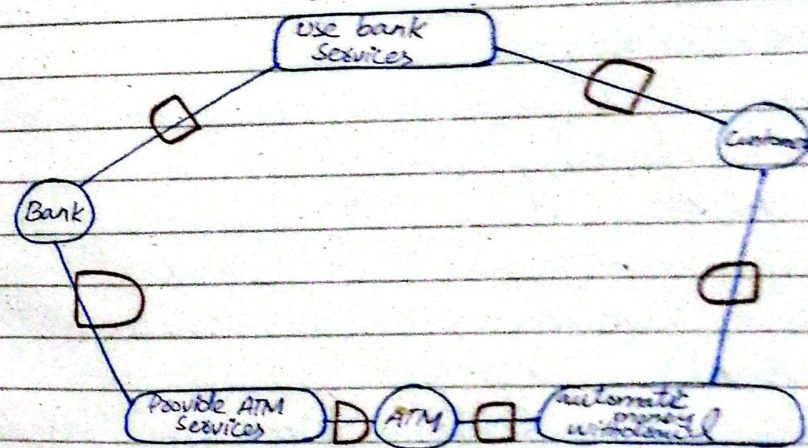
→ This diagram shows the different stakeholders involved in the organization, both internal and external.

→ it also shows how these stakeholders interact with each other and the organization.

→ Actors in actor diagram should represent agents, roles and positions in organization.

Example

Actor diagram of Bank.



Stakeholders (Actors):

- Customer
- ATM
- Bank

Relationships and interactions:

- Customer actor depends on Bank actor to be able to use bank services.
- Customer actor also depends on the ATM actor, so that able to withdraw money.
- The Bank actor depends on the ATM actor, so that bank will be able to provide ATM services to its customers.

Goals:

- Customer:

To access bank services conveniently and efficiently.

- ATM:

Provide a reliable and secure way to access bank services. Bridge b/w customer & bank.

- Bank:

Satisfy customer needs by providing ATM services.

Rationale Diagrams:

→ The second step is constructing rationale diagram.

→ Goal of each stakeholder^{actor} is determine and rationale diagram is created for each actor/stakeholder in the organization.

→ The goal analysis phase is completed when all the relevant goals for each actor are specified and all the dependencies b/w them are identified.

Example

- The rationale diagram for a bank.

- Goals of bank:

- Gaining new customers
- Managing loans
- Providing services to customers

- ① The "gain new customers" goal for the bank is AND-decomposed into two smaller goals:

1. Search new customers

2. Enter new customers

- The "Enter new customers" goal is further AND-decomposed into three smaller goals:

1. Specify contract details
2. Enter customer info
3. Enable customer services

→ The "enter customer info" goal is then AND-decomposed into three even smaller goals:

1. Collect customer info
2. enter personal data
3. Enter financial situation

→ New dependency is discovered, the Bank actor depends on its customer in order to collect their data.

II

→ The "manage loans" goal is AND-decomposed into four smaller goals:

1. enter info
2. check financial situation
3. Agree on loan specifications
4. Grant a loan.

→ "Check financial situation" goal benefits from the enter "enter financial situation" goal.

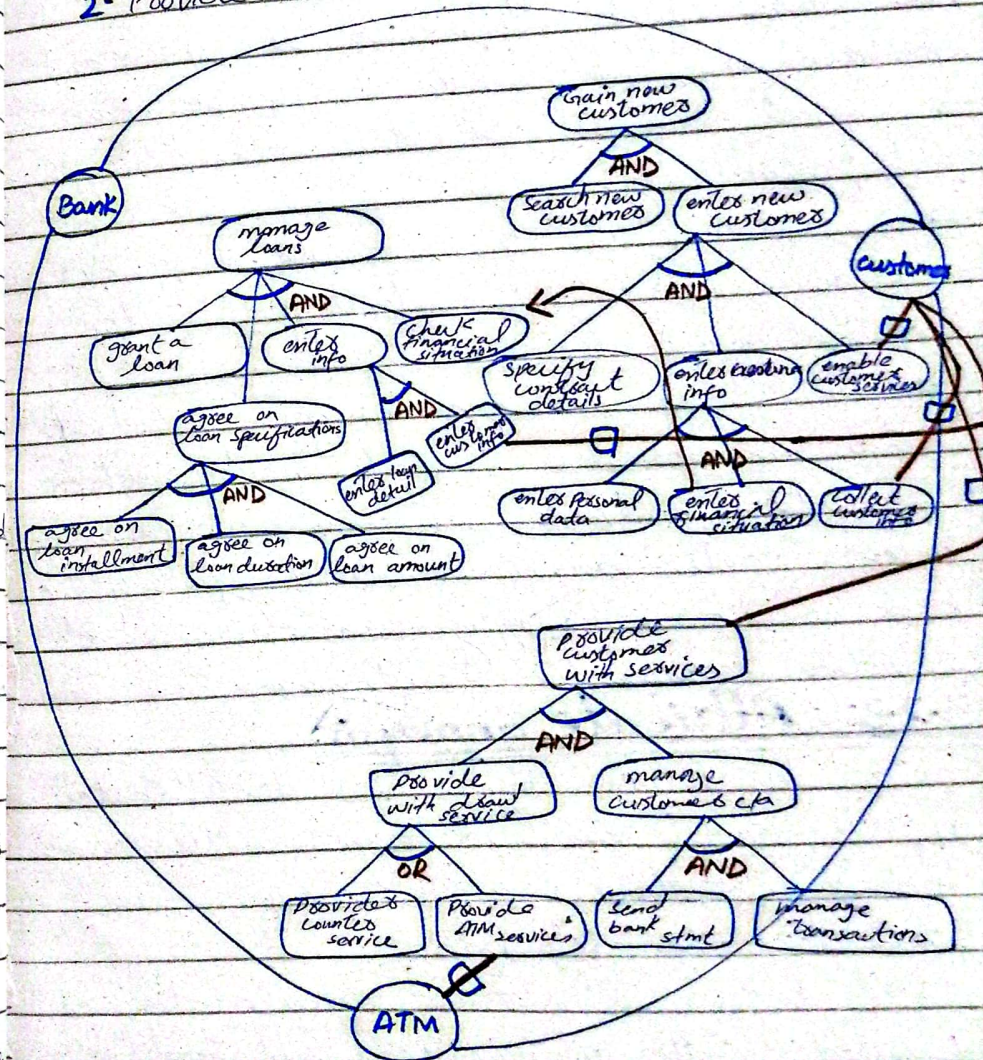
III

The "customer services" is AND-decomposed into:

- 1) Manage Customer cl
- 2) Provide withdrawal service

→ The customer service also OR-decomposed into:

1. Provide customer service
2. Provide ATM service.



3.2.2: Fact analysis:-

- In this phase, identify all those facts relevant to the organization.
- Then these facts links to the goals and dependencies identified in the previous phase.
- Fact analysis should be done carefully and systematically.
- Exploze all the rationale diagrams and goals in order to identify all the relevant facts.
- In previous page example of rationale diagram for bank includes facts such as transactions, deposits, withdrawals and payments.
- These facts are linked to the goals of the bank, such as managing transactions and providing services to customers.

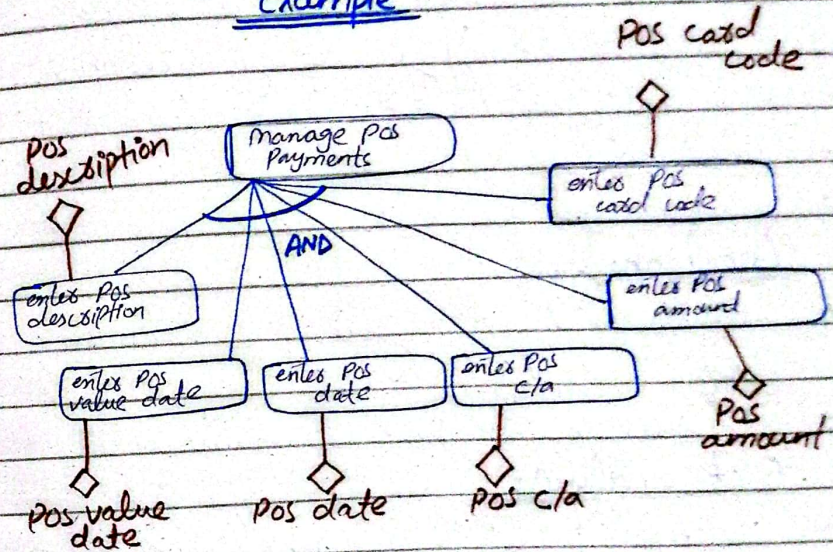
3.2.3: Attribute analysis:

In this phase, identify all the attributes of the facts that identified in the previous phase. Attributes are the specific details that describe the facts.

Review all the rationale diagrams created in the previous phase and should also

also explores the subgraphs of the goals to identify the attributes that are relevant to each goal.

Example



→ The attributes added to the "manage pos payments" goal are:

- pos value date
- pos date
- pos description
- pos c/a
- pos amount
- pos card code

3.3: Decision-making Modeling

→ Process used to understand and analyze how decisions are made within the organization.

→ It involves identifying the key decision-makers, their goals, and the factors that influence their choices.

→ it follows a four-step process:

Goal analysis:

- Understanding the goals of decision makers.

- Leads to rationale diagrams.

Fact analysis:

- Specifies facts or additional ^{rationale} information in order to create extended diagram.

Dimensional analysis:

- Adds dimensions to extended rationale diagrams.

Measure analysis:

- Adding measurements to extended rationale diagrams.

3.3.1: Goal analysis:

→ First step in decision making modeling process.

→ It involves:

- Examining actor diagrams - that show the roles of different people involved in decision-making.

- Identifying decision makers.

- Studying decision maker dependencies.

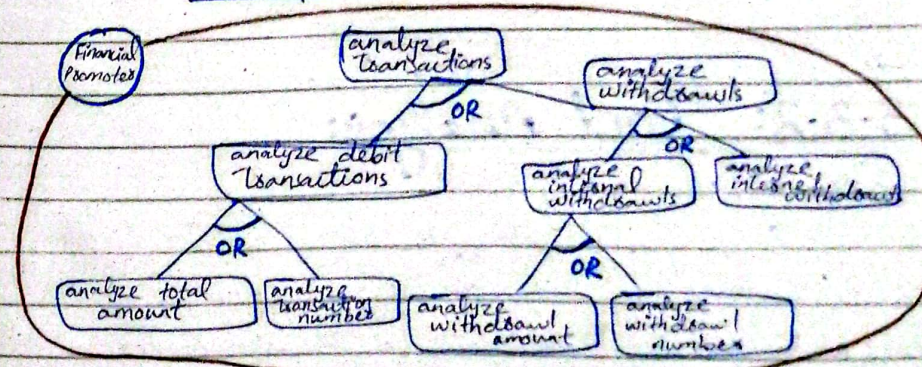
Goal decomposition:

→ Goals of each decision maker break down into smaller, more specific goals and studied them closely in order to draw a rationale diagram.

→ Goals can be completely different from those detected in the organizational modeling Phase.

→ They are integral part of the decision-making process.

Example:



→ This diagram is a rationale diagram for the financial promotes decision-maker.

→ Various main goals and have also subgoals of each goal.

Main Goal:

- Analyze transactions

Subgoals:

1. Analyze debit transactions.

- (i) Analyze total amount

- (ii) Analyze transaction numbers

2. Analyze withdrawals

- (i) Analyze internal WD

- Analyze withdrawal amount

- Analyze withdrawal numbers

- (ii) Analyze External WD

→ All goals are OR-decomposed.

3-3-2: Fact Analysis:

→ In fact analysis:-

Analysts should extend the rationale diagrams by identifying facts and match those facts to decision-maker goals.

→ Facts are usually imported from extended rationale diagrams of the organizational modeling phase.

Example:

→ Extended rationale diagram of financial promotes on next page.

→ Show the transactions part, which was detected in the organizational modeling phase to analyze transactions goal.

→ Sometimes analysts can also add new facts as they directly analyze rationale diagrams of decision-maker.

3-3-3: Dimensional Analysis:

→ In this phase, each fact is connected to those dimensions that decision-maker considers important for every decision-making to be reached.

Example:

→ Consider extended rationale diagram of financial promotes on next page.

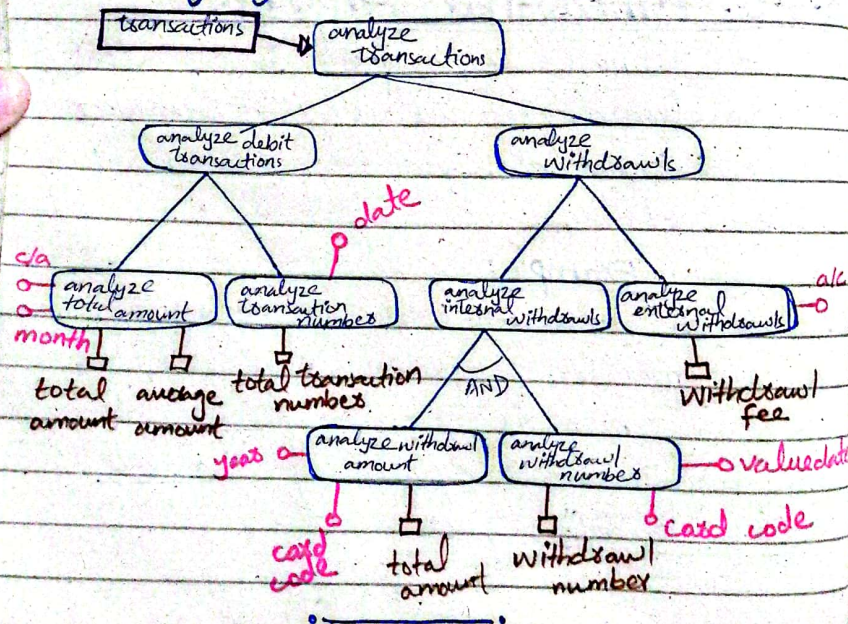
- the cta and month dimensions are specified to reach the "analyze total amount" goal.

3.3.4: Measure analysis:

→ In this phase, analysts should:
Connect a set of measures
to every fact previously identified.
Example:

→ The average amount and total amount
measures are identified for the "analyze
amount" fact.

Extended Rationale diagram of financial promoters:



4: Additional Requirements

→ Beyond the basic specifications of
facts and preliminary workload, some
additional specifications are essential for
various phases of data warehouse projects.
→ They should be collected from end
users and information system administrators.
→ Those specifications concern the following:

1. Logical and physical design:

• These requirements relate to how
the data warehouse will be structured
and implemented.

• Both logical and physical design are
influenced by the constraints posed by
data warehouse designers and corporate IT
staff.

→ Logical design refers to the conceptual
structure of the data warehouse, including the
entities, attributes and relationships between
them. It is like creating blueprint for
the warehouse.

→ Physical design refers to the actual
implementation of the data warehouse.

including the choice of hardware, software and database technologies. it is like building the warehouse based on the blueprint.

2. Data-staging design:

→ This refers to how data is initially stored and processed before being loaded into the data warehouse.

→ The frequency of data staging depends on the needs of end users and the capabilities of operational sources.

→ Frequency of data staging refers to how often data is extracted from operational systems and loaded into the staging area of a data warehouse.

→ Typically, frequency values of data staging range from a day to a week.

→ Special real-time applications can also receive hourly updates.

3. Data quality:

This refers to the accuracy and freshness of the data.

it is important to set criteria and thresholds for discarding faulty data and to establish ^{during ETL}

a reliable workflow for ensuring data quality.

4. Data warehouse architecture:

→ This refers to the overall structure and design of the data warehouse, including the:

- number of layers
- Use of data marts
- Reconciled layer materialization techniques

5. Data analysis applications:

→ This refers to the types of queries and reports that end users will need to perform on the data warehouse.

→ it is important to consider the features of the analysis tools that will be used.

6. Startup schedule:

→ This refers to the timeline for implementing the data warehouse.

→ Schedule for startup a data warehouse decided with support from DBA who provided their expertise and ensure their complete collaboration with end-users.

7. Training schedule:

- it is important to provide training to end users so that they can effectively use the data warehouse.
 - Designers should create a training schedule and offer follow-up sessions as needed.
-